vllm-project / **vllm** Public

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[core][distributed] add zmq fallback for broadcasting large objects #6183

New issue

Merged

youkaichao merged 11 commits into `vllm-project:main` from `youkaichao:add_zmq` on Jul 10, 2024

Conversation 8 | Commits 11 | Checks 0 | Files changed 6 | +274 −80

youkaichao commented on Jul 7, 2024 Member

The input to vision language model contains images, which has variable length and can be quite large.

While the shared memory broadcast introduced in #5399 works fine for LLMs, later we find we often need to adjust the buffer size for vision language models.

Estimating the size upper bound can be difficult. To solve the problem, this PR adds a fallback option using zeromq.

- When the object is small, we will use shared memory broadcast, which is fast and efficient.
- If the object is too large, we leave an overflow message in the shared memory, and send the data via zeromq, which can handle arbitrary sized data.

In addition, shared memory broadcast is limited to single node, while zeromq (socket-based) is not. Therefore, we can extend the broadcast to also work for cross-node settings. This PR extends the functionality.

cc @DarkLight1337 @ywang96 for vision language model related.

youkaichao added 9 commits 2 years ago

- `update in_the_same_node_as` 5da4157
- `add fallback to zmq for large data` 81e47ed
- `update tests` 61a4b82
- `bugfix` dcf1e6d

Reviewers

zhuohan123

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

4 participants

bugfix b8251d5

bugfix 0cf91d2

bugfix 710c9a3

rename to message queue 24e496d

update parallel state 45d82fc

youkaichao commented on Jul 7, 2024 View reviewed changes

vllm/distributed/device_communicators/shm_broadcast.py

Comment on lines +415 to +425

```
415 +        if self.n_local_reader > 0:
416 +            if len(serialized_obj) >
417 +                with self.acquire_wr
418 +                    buf[0] = 1  # ov
419 +                self.local_socket.se
420 +            else:
421 +                with self.acquire_wr
422 +                    buf[0] = 0  # no
423 +                    buf[1:len(serial
424 +        if self.n_remote_reader > 0
425 +            self.remote_socket.send(
```

youkaichao on Jul 7, 2024 Member Author

this is the most critical part, previously, when object size is too large (e.g. large image data), vLLM will error. Now, we will fall back to zqm.

DarkLight1337 commented on Jul 8, 2024 • edited Member

This idea sounds good, but I don't have much experience with cross-device communication, so I'll leave the review to someone who is more qualified.

zhuohan123 approved these changes on Jul 9, 2024 View reviewed changes

zhuohan123 left a comment Member

I took a very brief look and the code in general LGTM. I didn't check the internal logic of the message queue-based broadcast, but the interface and code change looks good. Please let me know whether I need to look into any part more carefully.

vllm/distributed/parallel_state.py Outdated Show resolved

youkaichao added 2 commits 2 years ago

rename daa49cc

update comments 17d7e36

WoosukKwon commented on Jul 10, 2024 Collaborator

@youkaichao `pytest tests/distributed/test_shm_broadcast.py` worked on the AMD MI210 machine. Do you want other kinds of tests as well?

youkaichao commented on Jul 10, 2024 Member Author

@WoosukKwon thanks for testing! I think it works because I added the dependency in `requirements-common.txt`, so all platforms can use it.

youkaichao merged commit **da78cae** into `vllm-project:main` on Jul 10, 2024

youkaichao deleted the `add_zmq` branch 2 years ago

adityagoel14 pushed a commit to adityagoel14/vllm-torchrun-test that referenced this pull request on Jul 11, 2024

[core][distributed] zmq fallback for broadcasting large objects (vllm… f7b7547

…

ywang96 mentioned this pull request on Jul 12, 2024

[Bug]: llava model gets stuck with RuntimeError: Please increase the max_chunk_bytes parameter. #6376 Closed

xjpang pushed a commit to xjpang/vllm that referenced this pull request on Jul 24, 2024

`[core][distributed] zmq fallback for broadcasting large objects (vllm…` ... `675d169`

mgoin mentioned this pull request on Jul 26, 2024

[RFC]: Isolate OpenAI Server Into Separate Process #6797 Closed

Alvant pushed a commit to compressa-ai/vllm that referenced this pull request on Oct 26, 2024

`[core][distributed] zmq fallback for broadcasting large objects (vllm…` ... `e402263`

LeiWang1999 pushed a commit to LeiWang1999/vllm-bitblas that referenced this pull request on Mar 26, 2025

`[core][distributed] zmq fallback for broadcasting large objects (vllm…` ... `9a3bd83`